

Parallel Lines, Transversals & Polygon Angles

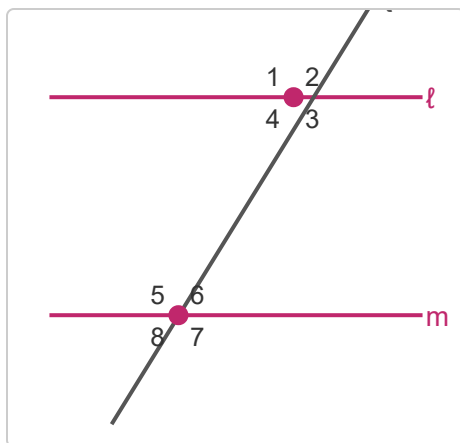
Student: Natalie Yoo

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Standard: CCSS.MATH.CONTENT.8.G.A.5 — Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles.

Part A — Name the Angle Pair (Q1–5)

Directions: In each diagram, lines ℓ and m are parallel, and line t is a transversal. Angles are labeled 1–8 as shown. Name the relationship between the two angles given. Choose from: corresponding, alternate interior, alternate exterior, co-interior (consecutive interior), vertical, linear pair.



Reference Diagram (use for Q1–5)

1. $\angle 3$ and $\angle 7$ Angle pair name: _____

2. $\angle 1$ and $\angle 5$ Angle pair name: _____

3. $\angle 4$ and $\angle 5$ Angle pair name: _____

4. $\angle 2$ and $\angle 6$ Angle pair name: _____

5. $\angle 1$ and $\angle 7$ Angle pair name: _____

Part B — Solve for x (Q6–10)

Directions: In each problem, lines $\ell \parallel m$ cut by transversal t . Use angle relationships to write an equation, then solve for x .

6. Corresponding angles: one angle measures 75° and the other measures $(11x - 2)^\circ$.

Solve for x . $x =$ _____

7. Alternate interior angles: one measures $(3x + 15)^\circ$ and the other measures 51° .

Solve for x . $x =$ _____

8. Co-interior (consecutive interior) angles add up to 180° . One angle is 110° and the other is $(7x + 5)^\circ$.

Solve for x . $x =$ _____

9. Alternate exterior angles: one measures $(5x - 10)^\circ$ and the other measures 60° .

Solve for x . $x =$ _____

10. Co-interior angles: one is $(4x + 12)^\circ$ and the other is $(2x + 24)^\circ$. Together they sum to 180° .

Solve for x . $x =$ _____

Part C — Quadrilaterals (Q11–13)

Directions: Answer each question about quadrilaterals. Remember: the four angles of any quadrilateral always add up to **360°**.

Name	Key Properties
Parallelogram	2 pairs of parallel sides; opposite sides equal; opposite angles equal
Rectangle	Parallelogram with 4 right angles
Rhombus	Parallelogram with 4 equal sides
Square	Rectangle + Rhombus (4 equal sides & 4 right angles)
Trapezoid	Exactly 1 pair of parallel sides
Kite	2 pairs of adjacent equal sides; no parallel sides

11. A quadrilateral has all four sides equal and all four angles equal to **90°**. What is its most specific name?

Name: _____

12. A quadrilateral has angles of **80°**, **100°**, and **80°**. Find the missing fourth angle.

Missing angle: _____

13. A quadrilateral has exactly one pair of parallel sides and its four angles are **65°**, **115°**, **65°**, and **115°**. What type of quadrilateral is it?

Type: _____

Part D — Polygon Basics (Q14–17)

Directions: Use the polygon name table and definitions below to answer each question.

Polygon names by number of sides:

3 = Triangle | 4 = Quadrilateral | 5 = Pentagon | 6 = Hexagon | 7 = Heptagon | 8 = Octagon

9 = Nonagon | 10 = Decagon | 11 = Hendecagon | 12 = Dodecagon | any $n = n$ -gon

Convex: all interior angles $< 180^\circ$ (no "dents").

Concave: at least one interior angle $> 180^\circ$ (has a "dent" or caves in).

Regular: all sides equal *and* all angles equal.

Irregular: sides or angles are not all equal.

14. A stop sign has 8 sides with all sides and angles equal. What is its name? Is it regular or irregular?

Name: _____ Regular / Irregular (circle one)

15. Circle **Yes** or **No**: Is a circle a polygon? _____

Explain in one sentence:

16. A shape has 10 sides but one interior angle is greater than 180° . Name the polygon and state whether it is convex or concave.

Name: _____ Convex / Concave (circle one)

17. A polygon has 12 sides, but its sides are not all the same length. What is it called?

Is it regular or irregular?

Name: _____ Regular / Irregular (circle one)

Part E — Interior Angle Sum (Q18–20)

Directions: Use the interior angle sum formula to answer each question.

$$\text{Sum of interior angles of a polygon with } n \text{ sides} = (n - 2) \times 180^\circ$$

- 18.** Find the sum of the interior angles of an **11-gon** (hendecagon).

Sum = _____

- 19.** Find the sum of the interior angles of a **25-gon**.

Sum = _____

- 20.** A **regular octagon** has 8 equal sides and 8 equal angles. What is the measure of *each* interior angle?

Hint: First find the total sum, then divide by 8.

Each angle = _____
